

MAYUR NARWADE

AI ENGINEER | GENERATIVE AI & MACHINE LEARNING | HEALTHCARE AI

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PROFESSIONAL SUMMARY

AI Engineer specializing in Generative AI, Machine Learning, Artificial Intelligence, healthcare AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Natural Language Processing (NLP), semantic search, vector databases, prompt engineering, and AI application deployment. Built a Dockerized healthcare RAG assistant using Llama 3.2, Ollama, ChromaDB, Sentence Transformers, FastAPI, Streamlit, evidence gating, source attribution, agentic routing, automated testing, and evaluation. Experienced in Python, PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, REST APIs, databases, Google Cloud Platform, data privacy, and technical documentation.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, SQL, HTML, CSS

Generative AI & NLP: RAG, LLM Integration, Prompt Engineering, Semantic Search, Text Embeddings, Sentence Transformers, Hugging Face Transformers, Ollama, Llama 3.2, ClinicalBERT, scispaCy

AI/ML: Artificial Intelligence, Machine Learning, Deep Learning, NLP, Computer Vision, CNN, TensorFlow, PyTorch, OpenCV, XGBoost

Backend & Data: FastAPI, Flask, REST APIs, ChromaDB, Vector Databases, PostgreSQL, MongoDB, MySQL

Deployment & Engineering: Docker, Docker Compose, Streamlit User Interfaces, AI Application Deployment, Git, GitHub, Google Cloud Platform, Vertex AI, Testing, Evaluation, Logging, Error Handling, Data Privacy, Clean Coding, Technical Documentation

EXPERIENCE

Technical Intern — National Institute of Electronics & Information Technology (NIELIT)

Chhatrapati Sambhajnagar | Feb 2026 – Present

- Contributed to healthcare AI development, backend integration, technical research, testing, documentation, and workflow improvement in a government R&D environment.
- Supported FastAPI services, REST API implementation, debugging, database workflows, dataset preparation, validation, data privacy, and technical documentation.
- Mentored **90+ students** and conducted **5+ technical sessions** during Advanced IoT and Microcontrollers bootcamps.

PROJECTS

Healthcare AI Assistant Using RAG and Local LLMs

Python | FastAPI | Llama 3.2 | Ollama | ChromaDB | Sentence Transformers | Streamlit | Docker

- Developed an end-to-end healthcare RAG pipeline that ingests synthetic policy documents, creates embeddings, performs semantic retrieval, and generates context-grounded answers using a locally hosted Llama 3.2 3B model.
- Processed **6 healthcare documents into 9 searchable chunks** and implemented relevance-based evidence gating, source citations, confidence scoring, and safe fallback behavior to reduce unsupported generation.
- Built FastAPI **/health**, **/ingest**, and **/ask** endpoints, a Streamlit user interface, and an agentic router with a simulated appointment-availability tool.
- Containerized FastAPI, Streamlit, and Ollama with Docker Compose; achieved **4/4 automated tests passed, 100% route accuracy, 100% citation match, and zero evaluation errors** on the included synthetic prototype evaluation set.

Repository: github.com/MayurNarwade/healthcare-ai-assistant-rag

Medication Error Detection System (MEDS)

Python | FastAPI | ClinicalBERT | scispaCy | OCR | XGBoost | RxNorm | PostgreSQL | MongoDB

- Contributed to an AI-assisted healthcare system for identifying potential medication errors from prescriptions, patient information, medicine knowledge, and clinical rules.
- Supported a hybrid pipeline combining OCR, clinical NLP, RxNorm normalization, rule-based validation, and XGBoost inference for interaction, allergy, dosage, and duplicate-medication risk analysis.
- Assisted with REST API development, database integration, testing, debugging, role-based access control, JWT authentication, data privacy, and technical documentation.

Driver Drowsiness Detection System (DDS)

Python | OpenCV | Computer Vision | Facial Landmarks | NVIDIA Jetson Nano

- Developed and tested a real-time computer-vision pipeline that monitors eye behavior and prolonged eye closure to detect potential driver drowsiness.
- Applied temporal thresholds to distinguish normal blinking from sustained fatigue and integrated an automated alert mechanism.
- Optimized edge-device testing to achieve facial-landmark processing below **45 ms** while sustaining up to **30 FPS**.

EDUCATION

Bachelor of Technology in Computer Science and Engineering — MGM College of Engineering & Technology, Nanded | Nov 2022 – Jul 2026 |

CGPA: 7.5/10.0

CERTIFICATIONS

Google Cloud: Inspect Rich Documents with Gemini | Multimodal RAG | Prompt Design in Vertex AI | Machine Learning I — Core Concepts

PUBLICATION

Zero-Shot Learning Based Multilingual Medical Symptom Understanding and Disease Inference System Using NLP and Semantic AI

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